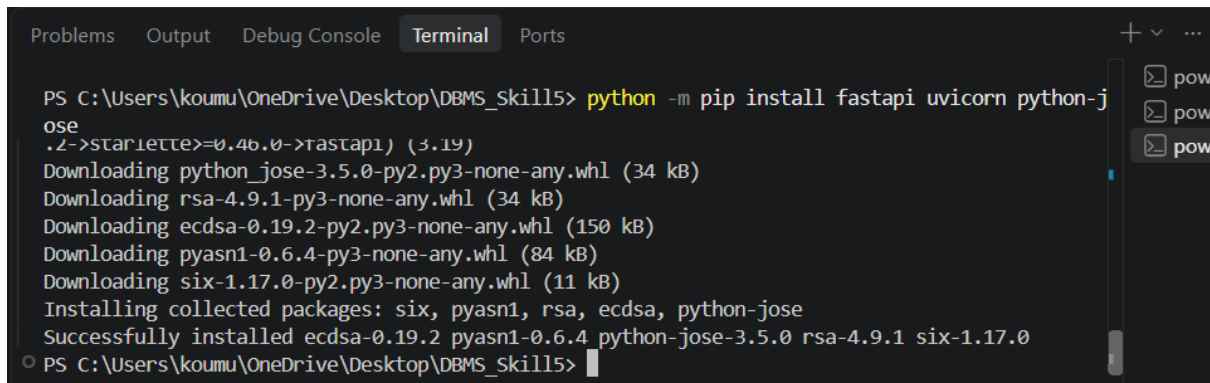


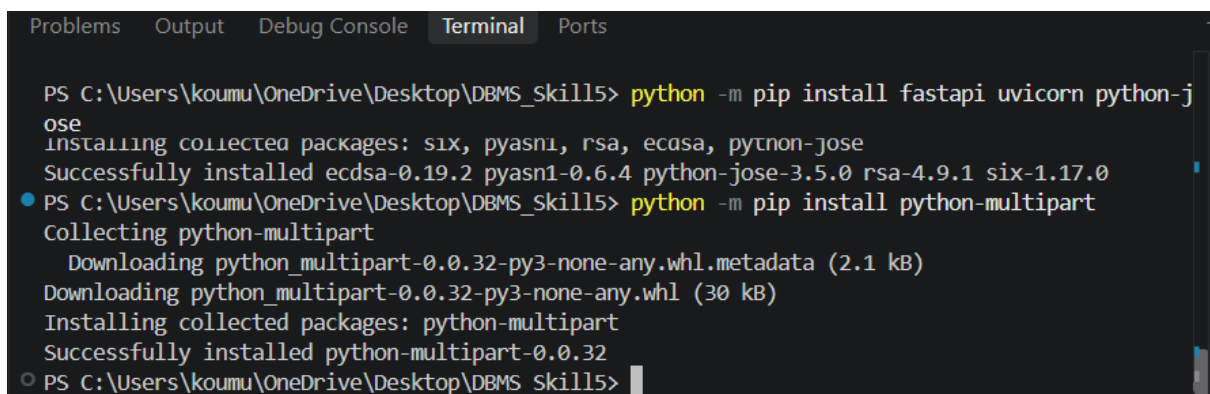
SKILL WEEK 6(DBMS)

Installation of fast api uvicorn:



```
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill5> python -m pip install fastapi uvicorn python-jose
Downloading python_jose-3.5.0-py2.py3-none-any.whl (34 kB)
Downloading rsa-4.9.1-py3-none-any.whl (34 kB)
Downloading ecdsa-0.19.2-py2.py3-none-any.whl (150 kB)
Downloading pyasn1-0.6.4-py3-none-any.whl (84 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, pyasn1, rsa, ecdsa, python-jose
Successfully installed ecdsa-0.19.2 pyasn1-0.6.4 python-jose-3.5.0 rsa-4.9.1 six-1.17.0
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill5>
```

Installation of python-multipart:



```
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill5> python -m pip install fastapi uvicorn python-jose
Installing collected packages: six, pyasn1, rsa, ecdsa, python-jose
Successfully installed ecdsa-0.19.2 pyasn1-0.6.4 python-jose-3.5.0 rsa-4.9.1 six-1.17.0
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill5> python -m pip install python-multipart
Collecting python-multipart
  Downloading python_multipart-0.0.32-py3-none-any.whl.metadata (2.1 kB)
  Downloading python_multipart-0.0.32-py3-none-any.whl (30 kB)
Installing collected packages: python-multipart
Successfully installed python-multipart-0.0.32
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill5>
```

```
main.py X
main.py
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15
16     # Classroom example only
17     if form_data.username == "101" and form_data.password == "abc123":
18
19         payload = {
20             "sub": form_data.username,
21             "role": "student"
22         }
23
24         token = jwt.encode(
25             payload,
26             SECRET_KEY,
27             algorithm=ALGORITHM
28         )
29
30         return {
31             "access_token": token,
32             "token_type": "bearer"
33         }
34
35     raise HTTPException(
36         status_code=401,
37         detail="Invalid username or password"
38     )
39
```

```
main.py X
main.py
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
38 )
39
40
41 def get_current_user(token: str = Depends(oauth2_scheme)):
42
43     try:
44         payload = jwt.decode(
45             token,
46             SECRET_KEY,
47             algorithms=[ALGORITHM]
48         )
49
50         return payload
51
52     except Exception:
53         raise HTTPException(
54             status_code=401,
55             detail="Invalid or expired token"
56         )
57
58
59 @app.get("/profile")
60 def profile(user=Depends(get_current_user)):
61
62     return {
63         "message": "Welcome to your profile",
64         "user_id": user["sub"],
65         "role": user["role"]
66     }
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\koumu\OneDrive\Desktop\DBMS_Skill15> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\koumu\\OneDrive\\Desktop\\DBMS_Skill15']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [2940] using StatReload
INFO: Started server process [15612]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: */*' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=101&password=abc123&scope=&client_id=string&client_secret=*****'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code Details

200

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWI0IjI0IiwiaWF0IjoiY3R1ZGVudCJ9.NHqwkDLPB1Q5BFQ1iKmdoWvLOjblUMdHm4aRQ6MK1zI",
  "token_type": "bearer"
}
```

Response headers

```
content-length: 162
content-type: application/json
date: Thu, 17 Sep 2026 17:30:14 GMT
server: uvicorn
```

GET /profile Profile

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWI0IjI0IiwiaWF0IjoiY3R1ZGVudCJ9.NHqwkDLPB1Q5BFQ1iKmdoWvLOjblUMdHm4aRQ6MK1zI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code Details

200

Response body

```
{
  "message": "Welcome to your profile",
  "user_id": "101",
  "role": "student"
}
```

Response headers

```
content-length: 70
content-type: application/json
date: Thu, 17 Sep 2026 17:36:13 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links
Media type		
application/json		
Controls Accept header.		
Example Value Schema		
"string"		